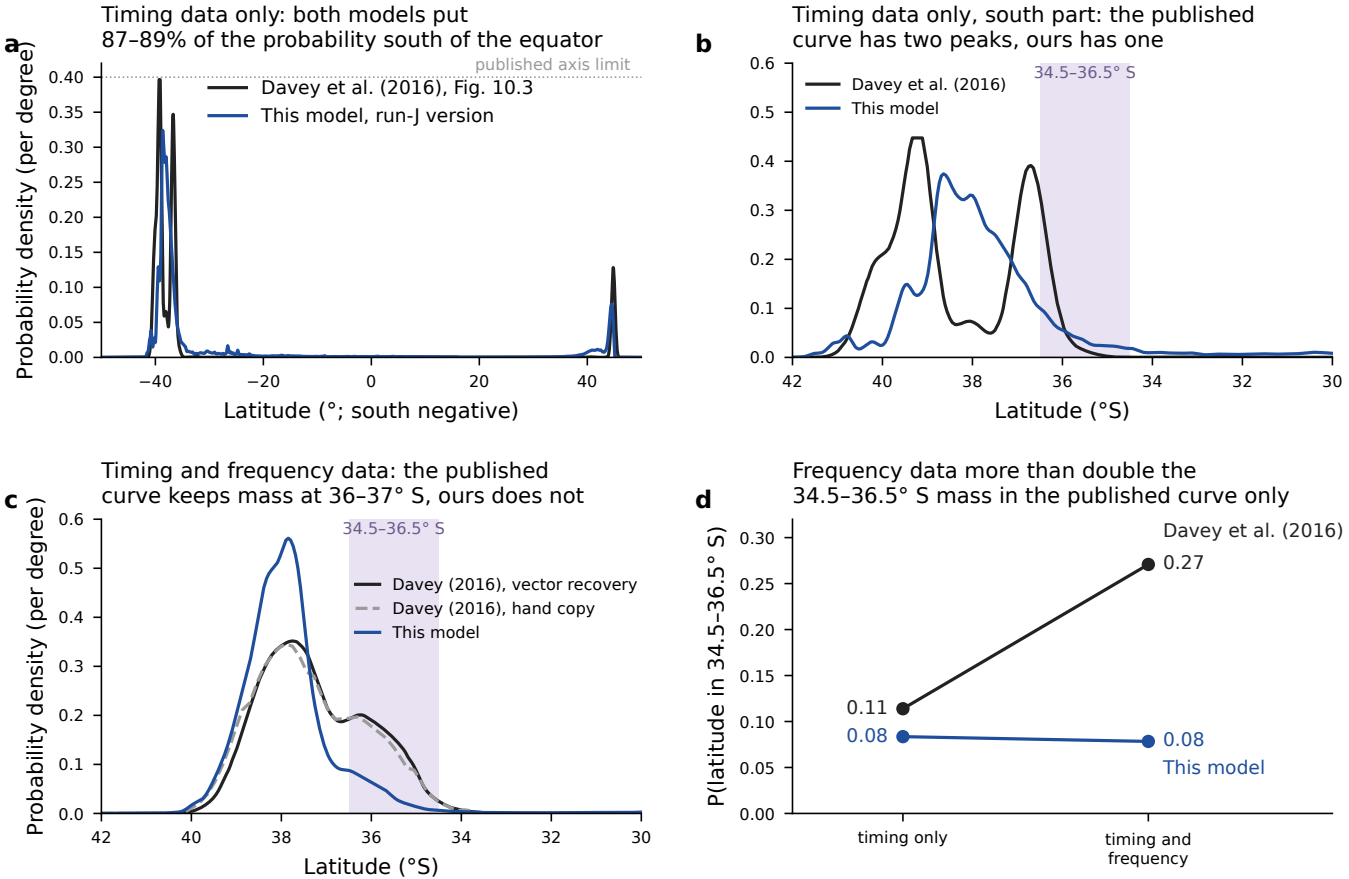

Plain: These charts compare where the aircraft crossed the last arc in the Davey et al. (2016) book and in our model, which is set up to copy the book. Charts a and b use only the signal timing data. Charts c and d also use the signal frequency data. The book curves come from the drawing data in the book PDF; they are not traced by hand. Without frequency data, the two models agree on the north-south split. In the south they do not agree: the book has two peaks (39.2° S and 36.7° S) and we have one peak (38.7° S). With frequency data, the book curve moves 1.4° north and keeps much mass at 36–37° S. Our curve does not move. Thus most of the difference comes from how the frequency data are used.

Technical: Davey et al. (2016) Fig. 10.3, top and bottom panels, recovered from PDF vector paths (pymupdf; Bézier segments sampled at 12 points; tick-mark calibration); the top-panel tip is clipped at the 0.4 axis limit (39.34–39.11° S, about 0.006 of area). Our curves: pure reproduction configuration davey2016-inmarsat.toml at tag run-j (1244b7c), Inmarsat satellite states, ERA5 winds, 5 steering modes, 7M particles per seed, 0.1° kernel smoothing. (a, b) BTO only, 8 seeds, built on the Mac (18 threads, 25 min), ln Z −66.19, split-half 0.907. (c) BTO + BFO, σ_BFO 7 Hz, 4 seeds (step-size check baseline). Panels b–c renormalised over latitudes south of 0°. South-renormalised medians: Davey 38.88 → 37.43° S, this model 37.91 → 37.95° S; 34.5–36.5° S mass: Davey 0.114 → 0.271, this model 0.084 (seed SE 0.004) → 0.078. Band masses use the exact interval on a 0.001° grid; the earlier hand copy of the bottom panel gives 0.250 (previously quoted 0.2406).